

Figure 3: ANDROID_HOME environment variable

Figure 4: Source code management

Figure 5: Lint configuration

- Create a pipeline so that the first code analysis is performed and on its successful implementation, execute another build job to create an APK file. Now, let's perform each step in sequence.

Configuring Git, Java, and Gradle: To execute the build pipeline, it is necessary to take code from a shared repository. As shown below, Git is configured to go further. The same configuration is applied to all version control that is to be set up. The path in Jenkins is *Home > Global tool configuration > Version control / Git*.

Figure 6: Publish Android Lint results

In an Android project, the main part is Gradle, which is used to build our source code and download all the necessary dependencies required for the project. In the *name* field, users can enter Gradle with their version for better readability. The next field is *Gradle Home*, which is the same as the environment variable in our system. Copy your path to Gradle and paste it here. There is one more option, 'Install automatically', which installs Gradle's latest version if the user does not have it.

Configuring the ANDROID_HOME environment variable: The next step is to configure the SDK for the Android project that contains all the platform tools and other tools also.

Here, the user has to follow a path to the SDK file, which is present in the system.

The path in Jenkins is *Home > Configuration > SDK*. **Creating a Freestyle project to perform Lint analysis for the Android application:** The basic setup is ready; so let's start our project. The first step is to enter a proper name (AndroidApp-CA) to your project. Then select 'Freestyle project' under *Category*, and click on OK. Your project file structure is ready to use.

The user can customise all the configuration steps to show a neat and clean function. As shown in Figure 4, in a general configuration, the 'Discard old build' option discards all your old builds and keeps the number of the build at whatever the user wants. The path in Jenkins is *Home > #your_project# > General Setting*.

In the last step, we configure Git as version control to pull the latest code for the Build Pipeline. Select the *Git* option and provide the repository's URL and its credentials. Users can also mention from which branch they want to take the code, and as shown in Figure 5, the 'Master' branch is applied to it. Then click the *Apply and Save* button to save all your configuration steps.

The next step is to add Gradle to the build, as well as add Lint to do static code analysis. Lint is a tool that performs code analysis for Android applications, just as Sonarqube does in Java applications. To add the Lint task to the configuration, the user has to write Lint options in the